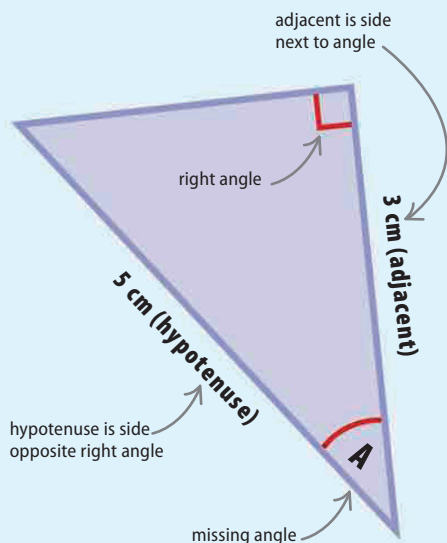


Using the cosine formula

In this right triangle the hypotenuse and the side adjacent to angle A are known. Use the cosine formula to find the size of angle A.



Choose the right formula. In this example the hypotenuse and the side adjacent to the missing angle, A, are known, so use the cosine formula.

Substitute the known values into the formula.

Work out the value of cos A by dividing the adjacent side by the length of the hypotenuse.

Find the value of the angle by using the inverse cosine function on a calculator.

Round the answer to a suitable size. This is the value of the missing angle.

$$\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\cos A = \frac{3}{5}$$

$$\cos A = 0.6$$

$$A = \cos^{-1}(0.6)$$

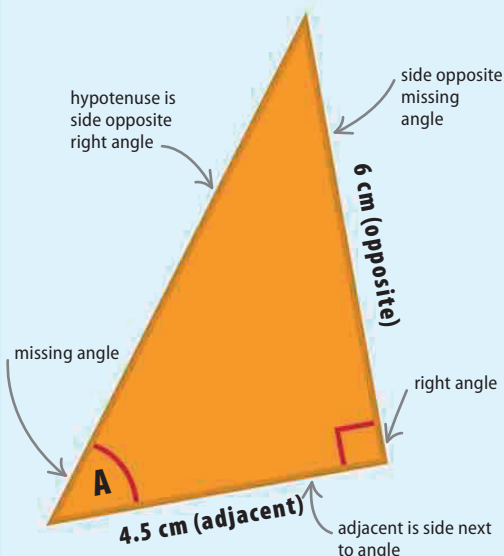
press shift then cosine button to get inverse cosine

$$A = 53.13^\circ$$

answer is rounded to 2 decimal places

Using the tangent formula

In this right triangle the sides opposite and adjacent to angle A are known. Use the tangent formula to find the size of angle A.



Choose the right formula—here the sides opposite and adjacent to the missing angle, A, are known, so use the tangent formula.

Substitute the known values into the tangent formula.

Work out the value of tan A by dividing the opposite by the adjacent.

Find the value of the angle by using the inverse tangent function on a calculator.

Round the answer to a suitable size. This is the value of the missing angle.

$$\tan A = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan A = \frac{6}{4.5}$$

$$\tan A = 1.3$$

$$A = \tan^{-1}(1.3)$$

answer rounded to 1 decimal place

press shift then tangent button to get inverse tangent

$$A = 52.43^\circ$$

answer is rounded to 2 decimal places